



(Year 2 / A2 Level)

Alternating Currents Lecture Notes

edited by Learmates

originally made by *Yuhao Yang*

CHAPTER 21

Alternating Currents

a **direct current** (d.c.) flows in one direction only

an **alternating current** (a.c.) reverses its flow direction from time to time

a.c. has certain advantages than d.c., as you will see in this chapter

we will study the mathematics of a.c. and the transmission process for a.c.

21.1 sinusoidal a.c.

as we have seen in §??, currents produced from generators are naturally sinusoidal

sinusoidal a.c. is one of the most important types of a.c. waveform in electrical engineering.

for most cases in this course, we focus on a.c. that varies like a sine wave

21.1.1 sinusoidal waveform

current: $I = I_0 \sin \omega t$ / voltage: $V = V_0 \sin \omega t$

➤ I_0, V_0 are called *peak current* and *peak voltage* (amplitudes of the a.c. signal)

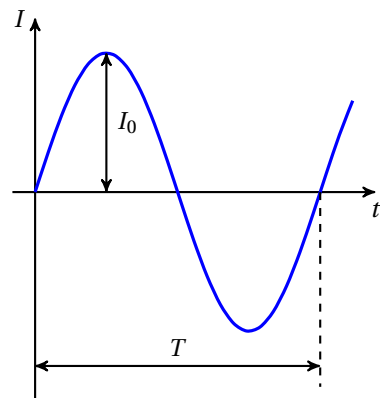
➤ ω is **angular frequency**, which describes how fast a.c. signal oscillates (same idea as for simple harmonic motion, see §??)

frequency and period of the signal are given by:

$$\omega = 2\pi f, \quad T = \frac{1}{f} = \frac{2\pi}{\omega}$$

➤ mean current: $\langle I \rangle = 0$, mean voltage: $\langle V \rangle = 0$

this is because a.c. signal fluctuates with time, positive and negative bits cancel out



Question 21.1 An alternating voltage has a peak value of 16 V and period of 0.10 s. Write down a mathematical equation that describes the variation of this voltage.

Question 21.2 An alternating voltage is produced from a simple generator. If the rotating speed of the coil in the generator doubles, describe quantitatively the change in the peak value and frequency of the alternating voltage.

Question 21.3 A student argues that when an alternating current is driven through a resistor, the mean current is zero, so an alternating current does not produce heating power on the resistor. State and explain whether this is correct.

21.1.2 power

electrical power dissipated in a resistor: $P = I^2 R$, or, $P = \frac{V^2}{R}$

since $I^2, V^2 \geq 0$, P can never be negative, so an a.c. can produce effective power in a resistor

note that for an a.c., P keeps changing with time, as I and V are both varying with time

in everyday life, we are more concerned about the *mean power*

mean power output for an a.c. is: $\langle P \rangle = \langle I^2 \rangle R = \frac{\langle V^2 \rangle}{R}$

so we see the necessity to introduce mean square values $\langle I^2 \rangle$ and $\langle V^2 \rangle$

let's further introduce root mean square (r.m.s.) values: $I_{\text{rms}} = \sqrt{\langle I^2 \rangle}$, and $V_{\text{rms}} = \sqrt{\langle V^2 \rangle}$

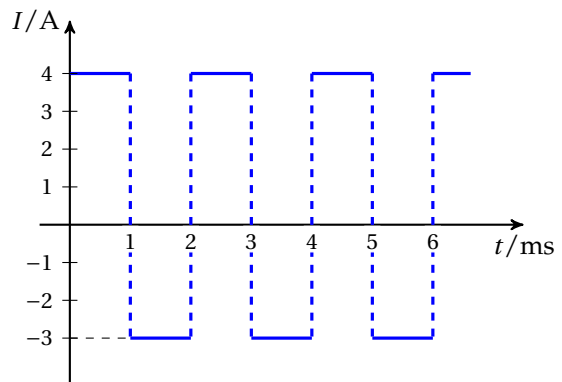
we can now write the mean power for a.c. as: $\langle P \rangle = I_{\text{rms}}^2 R = \frac{V_{\text{rms}}^2}{R}$

Example 21.1 The variation with time of an alternating current in a resistor of 120Ω is shown. (a) What is the value of its r.m.s. current? (b) What is the mean power dissipated by the resistor?

$$\langle I^2 \rangle = \frac{4^2 + (-3)^2}{2} = 12.5 \text{ A}^2$$

$$I_{\text{rms}} = \sqrt{\langle I^2 \rangle} = \sqrt{12.5} \approx 3.54 \text{ A}$$

$$\langle P \rangle = I_{\text{rms}}^2 R = 12.5 \times 120 = 1500 \text{ W} \quad \square$$



21.1.3 r.m.s. current & r.m.s. voltage

in last section, we studied the mathematical aspect of the r.m.s. value

but we still need a definition for r.m.s. current and r.m.s. voltage from a physical viewpoint

physically, r.m.s. value of an alternating current is defined as follows

r.m.s. current/voltage of an a.c. equals a steady d.c. current/voltage that delivers same average power to a resistive load

➤ for *sine waves*, r.m.s values are related to peak values by: $I_{\text{r.m.s}} = \frac{1}{\sqrt{2}} I_0$ and $V_{\text{r.m.s}} = \frac{1}{\sqrt{2}} V_0$

proof (*): total energy dissipation in one period is: $W_T = \int_0^T P dt$

so mean power can be given by: $\langle P \rangle = \frac{W_T}{T} = \frac{1}{T} \int_0^T P dt$

substitute $P = I^2 R \stackrel{I=I_0 \sin \omega t}{=} I_0^2 R \sin^2 \omega t$, we have: $\langle P \rangle = \frac{I_0^2 R}{T} \int_0^T \sin^2 \omega t dt$

the integral is carried out: $\int_0^T \sin^2 \omega t dt = \frac{1}{2} \int_0^T (1 - \cos 2\omega t) dt = \frac{1}{2} \left(t - \frac{\sin 2\omega t}{2\omega} \right) \Big|_0^T = \frac{1}{2} T$

now we have: $\langle P \rangle = I_{\text{rms}}^2 R = \frac{1}{2} I_0^2 R \Rightarrow I_{\text{rms}} = \frac{1}{\sqrt{2}} I_0$

a similar calculation for voltage would show: $V_{\text{rms}} = \frac{1}{\sqrt{2}} V_0$ □

➤ it is worth pointing out that the $\frac{1}{\sqrt{2}}$ -relation for r.m.s. values only holds for sine waves

for other waveforms, e.g., square waves or triangle waves, numerical constant is different

➤ value of voltages stated for mains electricity supply usually refers to the r.m.s. value^[1]

Example 21.2 An a.c. power supply produces a sinusoidal output across a resistor of 30Ω .

The maximum voltage is found to be 75 V . Find energy dissipated in the resistor in 2.0 minutes.

✏ r.m.s voltage: $V_{\text{rms}} = \frac{1}{\sqrt{2}} V_0 = \frac{75}{\sqrt{2}} \approx 53.0 \text{ V}$

mean power output: $\langle P \rangle = \frac{V_{\text{rms}}^2}{R} = \frac{53.0^2}{30} \approx 93.8 \text{ W}$

energy dissipation: $W = \langle P \rangle t = 93.8 \times 2.0 \times 60 \approx 1.13 \times 10^4 \text{ J}$ □

^[1] Different countries have different standards. China mains electricity supplies a r.m.s. voltage of 220 V . UK uses a r.m.s. 230 V distribution system. USA has national standard of a r.m.s. 110 V voltage.

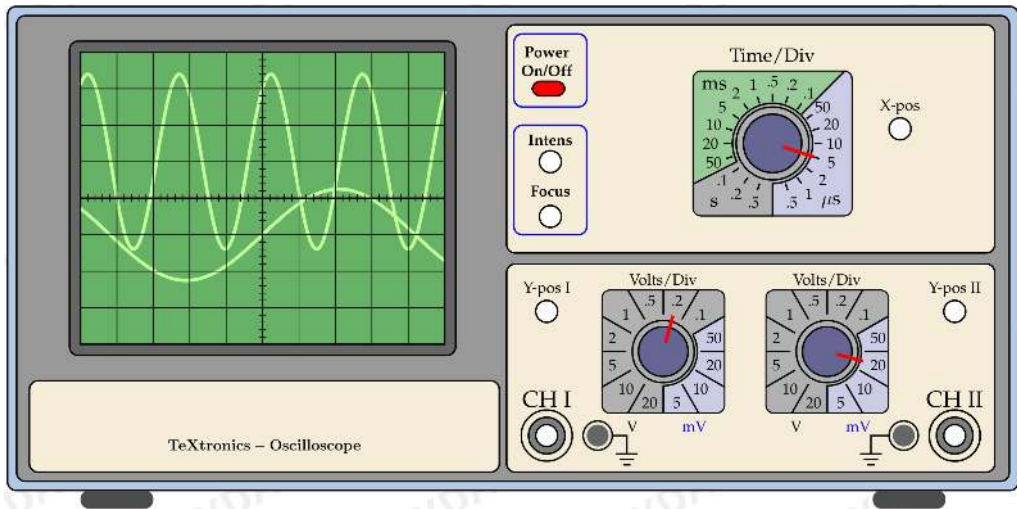
Question 21.4 An alternating voltage V is represented by the equation: $V = 310\sin(100\pi t)$, where V and t are measured in SI units. For this voltage, find (a) the peak voltage, (b) the r.m.s voltage, (c) the frequency, (d) the mean power when it is applied across a resistor of $250\ \Omega$.

Question 21.5 What is the average power dissipated in a resistor when the alternating supply has a peak current of 5.0 A and a peak voltage of 8.0 V ?

Question 21.6 The peak value of a sinusoidal alternating current is equal to a steady direct current. When they are applied to the same load, what is the ratio of power dissipation $\frac{P_{\text{d.c.}}}{P_{\text{a.c.}}}$?

21.1.4 measurement of a.c. with an oscilloscope

to measure a varying voltage, we can use a **cathode-ray oscilloscope** (c.r.o.) (see figure^[2])



an oscilloscope is basically an electron deflection tube that uses a beam of electrons to trace the input voltage as a function of time.

electron beam is controlled by two sets of parallel plates

horizontal plates control how fast beam moves across screen, giving a horizontal time axis

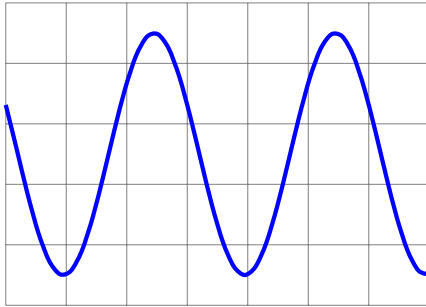
a.c. voltage is applied across vertical plates, causing beam to bend upwards or downwards

when beam hits fluorescent screen, entire trace of the beam can be seen across the screen

^[2]The illustration of the oscilloscope was created by *Hugues Vermeiren*. The source code is downloaded from TeXample: <http://www.texample.net/tikz/examples/textronics-oscilloscope/>.

- to take readings from oscilloscope, remember it displays a voltage against time graph
- **voltage gain**, or **Y-gain**, tells the number of volts per vertical division
 - **time base** gives the time unit per horizontal division

Example 21.3 An oscilloscope displays an a.c. voltage signal as shown.



suppose the time base setting is 10 ms/div, and the voltage gain is 5 V/div.

4 vertical divisions from highest to lowest, so

$$\text{peak voltage: } V_0 = \frac{1}{2} \times 4 \times 5 = 10 \text{ V}$$

3 horizontal divisions between peak to peak, so

$$\text{period: } T = 3 \times 10 = 30 \text{ ms}$$

$$\text{frequency } f = \frac{1}{T} = \frac{1}{30 \times 10^{-3}} \approx 33.3 \text{ Hz} \quad \square$$

Question 21.7 In Example 21.3, if the time-base setting is 5 ms/div and the voltage gain is 2 V/div, write down an equation that represents this alternating voltage.

21.2 power supply systems

21.2.1 high-voltage transmission

electricity is sent from power stations to consumers around the country

for long-distance transmission, we need minimise energy losses

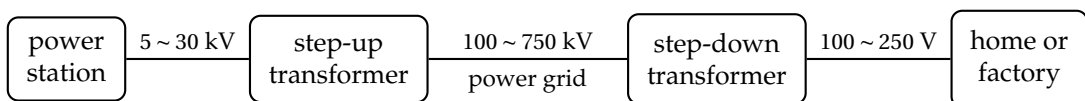
power dissipated due to resistance in cables is $I^2 R$, should transmit at low currents

since output power of a power station is fixed, low current means high voltage

so transmission at high voltages minimises energy loss in power grids

but for reasons of safety and efficiency, desirable to have low voltages at both generating end (power station) and receiving end (home or factory)

this requires converting a.c. into higher or lower voltages $\Rightarrow$ need for *transformers*

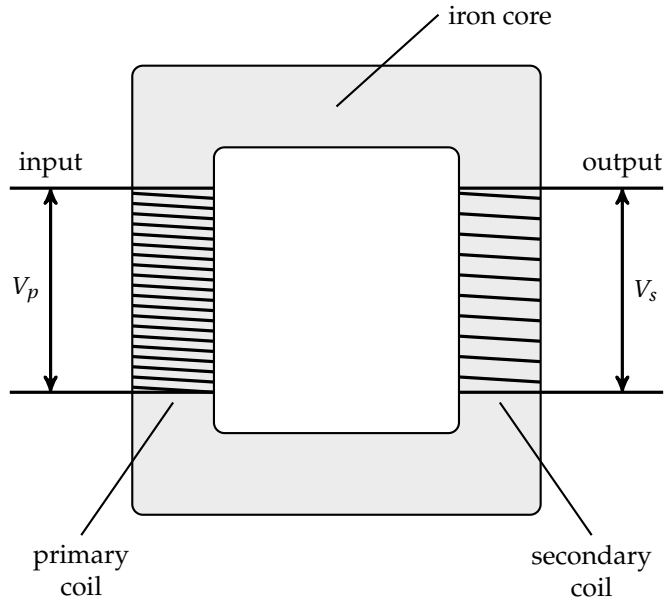


21.2.2 transformers

transformer is a device that can change values of voltage for an alternating current

transformers are an important part of the national power grid

transformers are essential for transmission, distribution, and utilization of electrical energy



structure of a typical transformer

➤ principle of transformers

a.c. flowing in **primary coil** (input) produces a changing magnetic field, i.e. a changing flux

this flux is linked with **secondary coil** through iron core

an a.c. voltage with same frequency is then induced in secondary coil (output)

➤ **turns-ratio equation** for a transformer

assume no loss in magnetic flux, i.e., all field inside transformer's iron core

from Faraday's law: $V_p = N_p \frac{d\phi}{dt}$, $V_s = N_s \frac{d\phi}{dt}$, where N_p , N_s are number of turns of coils

cancel out $\frac{d\phi}{dt}$, we find: $\frac{V_s}{V_p} = \frac{N_s}{N_p}$

if $N_p < N_s$, output voltage is increased → *step-up transformers*

if $N_p > N_s$, output voltage is decreased → *step-down transformers*

➤ if transformer is 100% efficient, then input power equals output power

for ideal transformers: $I_p V_p = I_s V_s$

➤ in practice, there always exists losses of energy from the transformer

causes of energy loss from the transformer include

- heat produced by *eddy currents* induced in iron core
this is reduced by *laminating* the core with insulate layers
- heat generated in coils due to resistance
can use thick copper wire to minimise resistance
- leakage of magnetic flux into surroundings

transformer's core is made of a continuous loop of iron to minimise this effect

Example 21.4 An ideal transformer has 200 turns on the primary coil and 5000 turns on the secondary coil. The r.m.s. input voltage to the primary coil is 8.0 V. What is the *peak* voltage across a resistor connected to the secondary coil?

$$\frac{V_{s,0}}{V_{p,0}} = \frac{N_s}{N_p} \Rightarrow \frac{V_{s,0}}{\sqrt{2}V_{p,\text{rms}}} = \frac{N_s}{N_p} \Rightarrow \frac{V_{s,0}}{8.0 \times \sqrt{2}} = \frac{5000}{200} \Rightarrow V_{s,0} \approx 2830 \text{ V} \quad \square$$

Question 21.8 An ideal transformer has 6000 turns on its primary coil. It converts a mains supply of 220 V r.m.s. to an a.c. voltage with a peak value of 12.0 V. Find the number of turns on the secondary coil.

Question 21.9 If a *steady* d.c. voltage is applied to the input of a simple transformer, what is the output voltage produced?

Question 21.10 State and explain whether the current in the primary coil of a transformer is in *phase* with the voltage induced in the secondary coil.

21.2.3 rectifiers

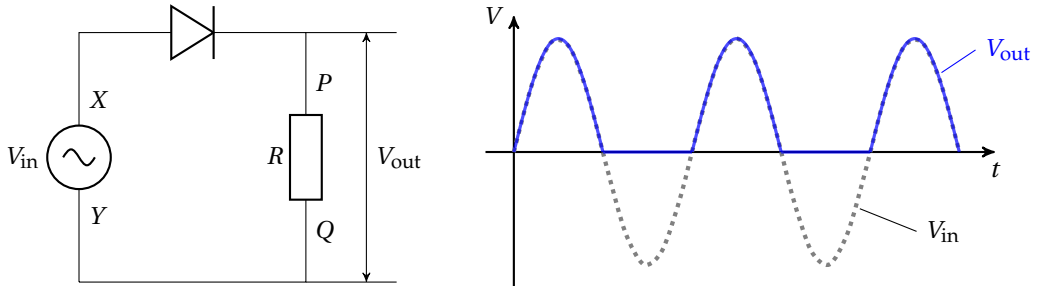
some electronic equipments (e.g., your smartphone, laptop, etc.) must work with d.c.

for these appliances require **rectification**, a process that converts an a.c. into a d.c.

rectification uses *diodes*, electronic components that only allow current in one direction

half-wave rectification

half-wave rectification uses a single diode



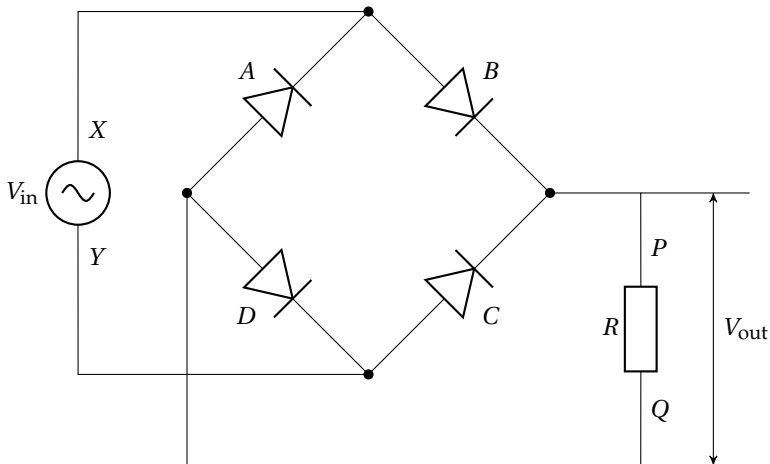
when input terminal X is positive, current can flow through diode, terminal P is positive with respect to Q for load resistor R

when input terminal Y is negative, flow of current is blocked, so zero output voltage on R
 V_{out} across load R is in one direction only, i.e., it becomes a d.c.

but power available from half-wave rectifier is only half of supply power

full-wave rectification

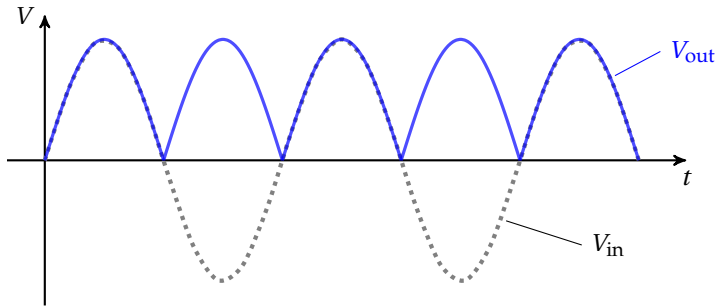
full-wave rectification requires using a combination of four-diode bridge structure



when input terminal X is positive, current flow: $X \rightarrow B \rightarrow R \rightarrow D \rightarrow Y$

when input terminal Y is positive, current flow: $Y \rightarrow C \rightarrow R \rightarrow A \rightarrow X$

either case, resulting output current in load resistor R always flows in same direction
 terminal P is always positive with respect to Q for R no matter what polarity for V_{in}
 output voltage across R is now full-wave rectified



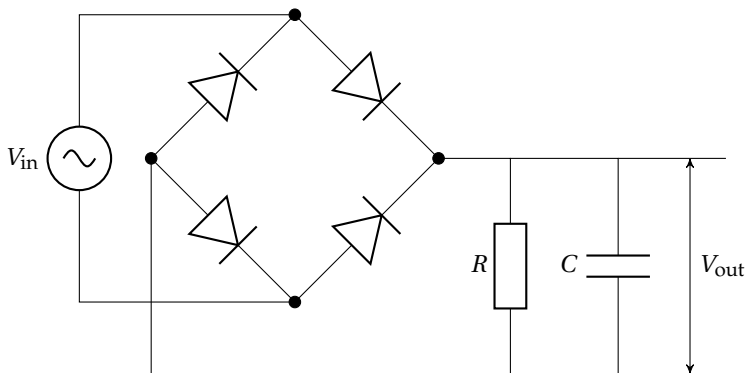
Question 21.11 If we want to have terminal Q to be positive with respect to P for load resistor R , how should we rebuild the bridge rectifier with the same four diodes?

21.2.4 smoothing

note that d.c. resulting from rectification still varies with time

to produce steady d.c. from fluctuating d.c., a process called **smoothing** is carried out

smoothing uses *capacitors*, which are connected in *parallel* with the load resistor (see figure)

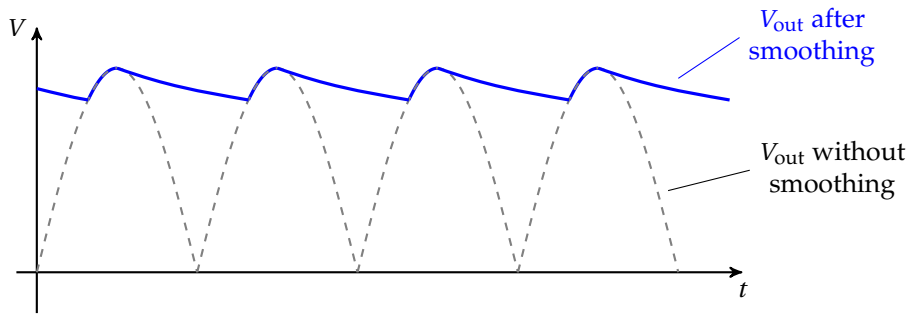


capacitor can store electrical energy as voltage output from rectifier rises

as voltage from rectifier drops, capacitor slowly discharges and feeds energy to the load

output voltage V_{out} across load will have less ripples over the cycles

less fluctuation in V_{out} so we say V_{out} is now smoothed



➤ effect of smoothing is controlled by choice of load resistor R and smoothing capacitor C

rate of charging and discharging depend on *time constant* RC (see §??)

greater RC , smoothing capacitor discharges more slowly, giving less ripples

a larger capacitor for a fixed resistor usually gives better smoothing effect

but if RC is too large, charging would become very slow, capacitor might not be completely charged after each cycle, this could also result into undesirable effects

Question 21.12 If a second capacitor is added in *series* with the first smoothing capacitor, use sketches to show the changes you would expect for the output voltage.